Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 May 2026
Team ID	HDI Predictor Team
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	4 Marks

Technical Architecture:

The HDI Predictor combines a Python notebook for analysis, a Flask backend for serving predictions, and HTML, CSS, and JavaScript for the browser interface.

The deliverable connects the model, data files, and frontend assets into one modular application.

Reference: HDI Predictor project context from the SmartBridge handouts.

S.No	Component	Description	Technology
1	User Interface	Prediction form, result page, and charts in the browser.	HTML, CSS, JavaScript, Chart.js
2.	Application Logic-1	Flask application factory and blueprint routing.	Python / Flask
1	Open-Source Frameworks	Core open-source libraries used in the project.	Flask, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
2	Security Implementations	Input checks and upload validation for user requests.	Python validation logic
3	Scalable Architecture	Modular notebook, model, route, and template structure.	Flask blueprints and reusable model files
4	Availability	Local Flask startup with a saved model artifact.	app.py and HDI.pkl
5	Performance	Fast single prediction flow and lightweight browser rendering.	Scikit-Learn model with client-side charts
8.	External API-1	Country auto-fill JSON endpoint.	Flask API

9.	External API-2	Batch prediction CSV upload route.	Flask upload route
10.	Machine Learning Model	Linear Regression prediction model.	Scikit-Learn
11.	Infrastructure (Server / Cloud)	Local development server for the browser demo.	Flask built-in server

Table-1 : Components & Technologies:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Core open-source libraries used in the project.	Flask, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn
2.	Security Implementations	Input checks and upload validation for user requests.	Python validation logic
3.	Scalable Architecture	Modular notebook, model, route, and template structure.	Flask blueprints and reusable model files
4.	Availability	Local Flask startup with a saved model artifact.	app.py and HDI.pkl

S.No	Characteristics	Description	Technology

5.	Performance	Fast single prediction flow and lightweight browser rendering.	Scikit-Learn model with client-side charts

Table-2: Application Characteristics:

References:

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d